Robust Secure Transmission for Satellite Communications

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In recent years, satellite communications have become the focus of research on space–air–ground–sea integrated wireless communication networks because of their wide coverage. Due to the broadcasting nature of satellite communications, the security of satellite systems has become a critical issue. In this article, we study the secure transmission of satellite communications, focusing on the geostationary earth orbit (GEO) and low earth orbit (LEO) satellite scenarios, respectively. Because of the satellite's high mobility and long propagation delays, the transmitter side cannot obtain accurate channel state information (CSI). To this end, we establish the channel phase uncertainty model of GEO satellite users, the channel angle uncertainty model of LEO satellite users, and the norm-bounded model of eavesdroppers, respectively. Then, a corresponding robust secure transmission design is proposed. Since the resulting optimization problems are nonconvex and challenging to handle, we propose an algorithm that transforms the original problem into a series of convex optimization subproblems and obtains the beamforming weight vectors through iterative calculation. Simulation results indicate that the security and robustness performance of the proposed algorithms are significantly improved compared with the conventional baselines.

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I. INTRODUCTION

In recent years, the space–air–ground–sea integrated wireless communication network supporting seamless, high-speed, and reliable transmission has attracted extensive attentions [1], [2]. Satellite communications have the superiority of wide-range coverage and can realize the global interconnection, thus becoming one of the hot spots in integrated network researches. The geostationary earth orbit (GEO) and low earth orbit (LEO) satellites are the main focuses of satellite communication researches. The GEO satellites with the multibeam architecture have spectrum multiplexing capabilities and high data transmission rates. They have been widely applied in civil and military fields, such as navigation, disaster relief, rescue, broadcasting, etc. [3], [4]. The LEO satellites have also gained significant attentions because of their relatively smaller path loss, less delay, and lower production costs than GEO ones [5], [6].

Because of the broadcast nature and colossal coverage area of wireless communications, satellite network information is vulnerable to eavesdropping, and security has become vital for satellite communications [7]. Traditionally, based on the assumption that eavesdroppers have limited computing capabilities, secure communications have consistently been implemented by the upper layer encryption technology of the protocol stack [8]. However, with the increasing processing ability, this assumption may become infeasible. Alternatively, by employing signal processing techniques and exploiting the inherent randomness of the wireless channel, physical layer security (PLS) provides a promising method for achieving secure communication, which has aroused widespread research interest [9], [10]. Compared with the terrestrial communication channels, the satellite location is higher and farther from the ground users and satellite communication signals in the transmission process have more interference, such as electromagnetic interference from the ground, interference from neighboring satellites, interference from climatic factors, and so on [11]. Therefore, satellite communications have longer propagation and processing delays. LEO satellites have high-speed mobility. In addition, the actual location information of eavesdroppers is challenging to obtain since they are not authenticated users [12]. Therefore, it is almost impossible for satellites to acquire accurate channel state information (CSI) of legitimate users and eavesdroppers [13], [14], [15], [16], and the traditional secure communication method of terrestrial communications, which requires perfect CSI of

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both the legitimate and eavesdropping links to calculate the secure capacity and design correspondingly wiretap coding, becomes not applicable for satellite communications. Moreover, compared with secure terrestrial communications, the satellite channel is more complex, the attenuation is more severe, and the communication distance is longer. Therefore, PLS transmission for satellites has more challenges and difficulties [17]. Motivated by this, we focus on the robust secure transmission for satellite communications in which the CSI of both the legitimate users and eavesdroppers are imperfectly known.

Related Works: Since the contributions of [7], [9], and [10], communication secrecy based on physical layer techniques has been widely studied. The authors in [18] studied the PLS problem in massive multiple-input–multiple-output (MIMO) multicast transmission exploiting the perfect CSI. In [19], the artificial noise (AN)-assisted jamming design of the Rician fading channel transmitter with large antenna array considering PLS was studied. In addition, the transmit beamforming optimization for providing PLS in a multibeam satellite system was studied in [20].

In the aforementioned works, nevertheless, the CSI of eavesdroppers was assumed wholly known at the transmit side. As a matter of fact, this assumption is unrealistic. In general, since eavesdroppers are usually keeping silent to hide, their accurate location information is hard to obtain. Moreover, even if eavesdroppers provide the channel estimation information to the satellites, due to some practical issues such as feedback errors and mismatches, the feedback results may be untrustworthy [21]. Motivated by these problems, in recent researches, imperfect CSI of eavesdroppers has been assumed [17], [22], [23], [24]. In [22], secure transmission design assisted by AN was investigated, in which the base station can acquire the statistical CSI of legitimate users and eavesdroppers. In [23], using imperfect CSI of eavesdroppers, the authors proposed a joint optimization algorithm of the power allocation and transmission covariance matrix under the global power constraint to realize the robust transmission design for multiple-input single-output (MISO) system. In [17], the robust secure transmission of multibeam satellites was addressed under the constraint of the total satellite transmission power and the assumption of imperfect CSI of the eavesdroppers. Moreover, the authors in [24] researched the problem of maximizing the total secrecy rate for satellite systems with the imperfect CSI of the eavesdroppers, in which each eavesdropper wiretap the corresponding legitimate user. In these studies, only the channel uncertainty of eavesdroppers is considered. Nevertheless, it is also challenging to acquire the perfect CSI of legitimate users in practical satellite communications since the systems have large transmission delays, and few studies take account of this situation. Motivated by this, we focus on the robust secure transmission design in satellite systems in which the CSI of both legitimate users and eavesdroppers are imperfect.

Contributions: In this article, robust transmission designs are investigated to maximize the secrecy rate and effectively improve the transmission security of satellite communication systems, especially GEO satellites and LEO satellites. In a nutshell, the main contributions of this article can be summarized as follows:

1) We utilize channel uncertainty models of legitimate users and eavesdroppers for GEO and LEO satellite systems, respectively. In GEO satellite systems, the channel phase uncertainty model of the user is considered because of the long propagation delay between the legitimate users and the gateway [15]. In addition, in LEO satellite systems, we utilize the channel angle uncertainty model of legitimate users caused by the high-speed movement of the satellite. Moreover, since the eavesdropper is not authenticated, it is impossible to know its specific location [17], [24], [25]. Therefore, we employ a norm-bounded model based on the position uncertainty of the eavesdropper [25].
2) Based on the channel uncertainty models, we investigate the robust sum secrecy rate maximization problems for GEO and LEO satellite systems under the total power constraints, respectively. Channel uncertainty in different communication links makes the original problem nonconvex and difficult to handle. To this end, we propose an algorithm framework to solve the problem effectively. The numerical analysis illustrates that the security performance of our proposed algorithm significantly outperforms the conventional baselines.

Organizations: The rest of this article is organized as follows. Sections II and III describe the system model, the robust secure transmission problem, and algorithm design in GEO and LEO satellite systems, respectively. In Section IV, the simulation and performance analysis of the proposed algorithm are carried out. Finally, Section V concludes this article.

Notations: In this article, we use bold uppercase and lowercase letters to denote matrices and vectors, respectively. The rest of the notations are shown in Table I for clarity.

II. ROBUST SECURE TRANSMISSION FOR THE GEO SATELLITE SYSTEM

Due to the high orbital altitude, large transmission delay, and many users served by GEO satellite systems, multigroup multicast communication systems are usually used to obtain a better system performance. Therefore, in this section, we consider the secure transmission of multigroup multicast GEO satellite communication systems.

A. Channel Model

In this section, we consider the downlink of the GEO multibeam satellite communications, where the system consists of a GEO satellite, N_u single antenna users, which are simultaneously served by N_t beams, and N_e eavesdroppers trying to intercept the sensitive data. In addition, we assume that full frequency reuse is employed among beams, and

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TABLE I
Notation Description

Notation	Description
$(\cdot)^T$	Transpose
$(\cdot)^H$	Hermitian conjugate
$\mathbb{C}^{m\times n}$	The set of complex matrices composed of m rows and n columns
$\odot$	Hadamard product
$\mathcal{N}(\mathbf{x},\mathbf{Y})$	The real-valued Gaussian distributions with mean vector $\mathbf{x}$ and covariance matrix $\mathbf{Y}$
$\mathcal{CN}(\mathbf{x},\mathbf{Y})$	The complex-valued Gaussian distributions with mean vector $\mathbf{x}$ and covariance matrix $\mathbf{Y}$
$\jmath$	Imaginary unit
$\exp\{\mathbf{x}\}$	The exponential function with e as the base and each element in vector $\mathbf{x}$ as the exponent
$\mathbb{E}\{\cdot\}$	The expectation operator
$\mathrm{diag}(\mathbf{x})$	The diagonal matrix with $\mathbf{x}$ along its principal diagonal elements
$\mathrm{Tr}(\cdot)$	The matrix trace operator
$\mathbf{X}\succeq\mathbf{0}$	$\mathbf{X}$ is positive semidefinite
$\|\mathbf{x}\|_2$	The ℓ_2-norm of $\mathbf{x}$
$[\mathbf{A}]_{m,n}$	The (m,n)th element of $\mathbf{A}$
$\triangleq$	Be defined as
$[x]^+$	$\mathrm{Max}(x,0)$
$\sim$	Be distributed as

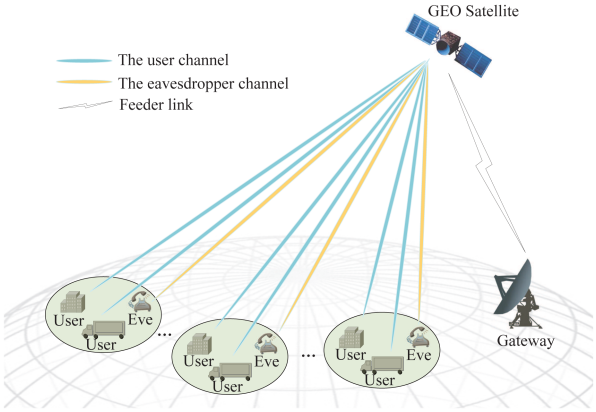


Fig. 1. Model of a downlink GEO multigroup multicast satellite communication system.

each beam has a multicast group to serve, so the system has a total of $K = N_\mathrm{t}$ multicast groups. Let $\mathcal{K} = \{1,\ldots,K\}$ represent the set of indices for all groups. Each user only belongs to one multicast group and the set of indices for users in the kth multicast group is denoted as $\mathcal{U}_k$ where $k \in \mathcal{K}$. In each multicast group, there exists one eavesdropper who aims to wiretap the information sent to this group, so the number of eavesdroppers satisfies $N_\mathrm{e} = K$. The system model of the GEO satellite is shown in Fig. 1.

1) *Channel Propagation Model:* In such a GEO satellite system, the downlink beam domain channel vector between the GEO satellite and the ith user in group k can be given by [26]

$$\mathbf{h}_i = \sqrt{\psi_i}\mathbf{t}_i^{\frac{1}{2}} \odot \mathbf{r}_i^{\frac{1}{2}} \odot \exp\{\jmath\boldsymbol{\theta}_i\}. \tag{1}$$

The coefficient ψ_i represents the free space loss (FSL), which is described as [11]

$$\psi_i = \frac{c}{4\pi f d_0} \tag{2}$$

where c represents the velocity of light, f describes the carrier frequency, and d_0 denotes the distance from the GEO satellite center to the ith user. In addition, $\mathbf{t}_i = [t_{i,1}, t_{i,2}, \ldots, t_{i,N_\mathrm{t}}]^T$ denotes the far-field beam pattern, and the component $t_{i,j}$, namely the beam gain from the jth beam to the ith user, can be approximated as [27], [28]

$$t_{i,j} = G_{r,i}\cdot G_T^j \cdot \left(\frac{J_1(u_{i,j})}{2u_{i,j}} + 36\frac{J_3(u_{i,j})}{u_{i,j}^3}\right)^2 \tag{3}$$

where $G_{r,i}$ and G_T^j are the receive antenna gain of the ith user and the transmit antenna gain for the jth beam, respectively. $J_1(\cdot)$ and $J_3(\cdot)$ denote the first kind Bessel functions of the first and third order, respectively. In addition, $u_{i,j}$ in (3) is represented as [27]

$$u_{i,j} = 2.07123\cdot\frac{\sin\varphi_{i,j}}{\sin\varphi_{j,3\mathrm{dB}}} \tag{4}$$

where $\varphi_{i,j}$ denotes the off-axis angle between the ith user and the center of the jth beam, and $\varphi_{j,3\mathrm{dB}}$ is the 3-dB angle for the jth beam. Furthermore, $\mathbf{r}_i = [r_{i,1}, r_{i,2}, \ldots, r_{i,N_\mathrm{t}}]^T$ denotes the rain coefficient[1] vector, which unit is decibel. Specifically, $\mathbf{r}_i^{\mathrm{dB}} = 20\log_{10}(\mathbf{r}_i)$, is commonly modeled as a lognormal random variable, i.e., $\ln(\mathbf{r}_i^{\mathrm{dB}}) \sim \mathcal{N}(\mu_r, \sigma_r^2)$ [26]. $\boldsymbol{\theta}_i = [\theta_{i,1}, \theta_{i,2}, \ldots, \theta_{i,N_\mathrm{t}}]^T$ is the channel phase vector with elements independently and uniformly distributed in $[0, 2\pi)$ [30], [31].

The $\exp\{\jmath\boldsymbol{\theta}_i\}$ in (1) represents the time-varying channel phases. It is worth noting that the scale parameter, rain attenuation, and far-field beam radiation pattern have an effect on the amplitude of the channel vector components. These coefficients remain constant over the coherent time interval [11]. Nonetheless, the channel phase is affected by time-varying components such as different local oscillators on the satellite, tropospheric fading, and low-noise block down-converters used at the user side. Furthermore, tropospheric fading has a much more significant effect on channel phase variation than amplitude [11], [32]. Therefore, for GEO systems, dealing with uncertainty in the channel phase is particularly importance.

2) *Channel Phase Uncertainty Model of Legitimate User:* We first describe the channel uncertainty model for legitimate users. The estimated channel vector $\hat{\mathbf{h}}_i$ is obtained by the ith user at instant t_0 and fed back to the gateway. Due to the propagation and processing delay, the actual

[1]In GEO satellite systems, signals will be affected by tropospheric effects during the transmission, such as atmospheric absorption, cloud fading, rain fading, and ionospheric scintillation, among which rain fading is the essential factor [29].

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channel vector $\mathbf{h}_i$ is employed at instant t_1 [33]. Since the locations of the legitimate users can be precisely known, the large-scale components in (1), i.e., the FSL, beam gain and rain fading coefficient vector, are considered to be known and fixed in the design. However, the channel phase $\boldsymbol{\theta}_i$ needs to be estimated. Hence, during the feedback interval, the amplitude remains constant and the phase is described as [32], [33]

$$\boldsymbol{\theta}_i(t_1) = \hat{\boldsymbol{\theta}}_i(t_0) + \mathbf{e}_i. \tag{5}$$

The long propagation delays cause the channel phase error which is denoted as $\mathbf{e}_i = [e_{i,1}, e_{i,2}, \ldots e_{i,N_t}]^T$. It is assumed that $\mathbf{e}_i$ follows the distribution of $\mathbf{e}_i \sim \mathcal{N}(\mathbf{0}, \sigma_i^2\mathbf{I})$ with the channel error variance σ_i^2 [11], [32]. Then, the relationship between $\mathbf{h}_i$ and $\hat{\mathbf{h}}_i$ is described as [13]

$$\mathbf{h}_i = \hat{\mathbf{h}}_i \odot \mathbf{q}_i = \mathrm{diag}(\hat{\mathbf{h}}_i)\mathbf{q}_i \tag{6}$$

where

$$\mathbf{q}_i \triangleq \exp\{\jmath\mathbf{e}_i\}. \tag{7}$$

We denote $\mathbf{Q}_i \triangleq \mathbf{q}_i\mathbf{q}_i^H$ and the correlation matrix of $\mathbf{q}_i$ can be represented as

$$\mathbf{A}_i = \mathbb{E}\{\mathbf{q}_i\mathbf{q}_i^H\} = \mathbb{E}\{\mathbf{Q}_i\}. \tag{8}$$

The elements of the matrix $\mathbf{A}_i$ can be written as [34]

$$[\mathbf{A}_i]_{m,n} = \begin{cases} 1, & m = n \\ \exp\left\{-\sigma_i^2\right\}, & m \neq n. \end{cases} \tag{9}$$

In practice, exact $\mathbf{e}_i$ is unknown to the designer, but σ_i^2 can be known via long-term measurement. Therefore, $\mathbf{A}_i$ defined in (8) will be exploited in our robust design.

Based on the aforementioned channel model, we can obtain the long-term channel correlation matrix of the legitimate user that will be used in the following section as

$$\begin{aligned} \mathbf{X}_i' &= \mathbb{E}\{\mathbf{X}_i\} = \mathrm{diag}(\mathbf{h}_i)\mathbb{E}\{\mathbf{Q}_i\}\mathrm{diag}(\mathbf{h}_i^H) \\ &= \mathrm{diag}(\mathbf{h}_i)\mathbf{A}_i\mathrm{diag}(\mathbf{h}_i^H) \end{aligned} \tag{10}$$

where $\mathbf{X}_i \in \mathbb{C}^{N_t \times N_t}$ represents the instantaneous channel correlation matrix. In addition, the expectation of the channel phase uncertainty is described as $\mathbf{A}_i$, which is previously defined in (9).

3) *Channel Uncertainty Model of Eavesdropper:* Different from the channel uncertainty model for legitimate users, for eavesdroppers, the locations are also unknown to the system designer due to their passive nature. In this case, CSI uncertainty exists in the entire channel, other than only existing in the phase component as previously assumed.

For this reason, we adopt a norm-bounded channel uncertainly model for eavesdroppers, which is commonly used in [17], [24], [25]. We assume that the transmit side only obtains the uncertainty regions of the eavesdroppers [25], which is described as

$$\Omega_e^{\mathrm{geo}} = \left\{\mathbf{h}_e \mid \mathbf{h}_e = \hat{\mathbf{h}}_e + \Delta\mathbf{h}_e, \|\Delta\mathbf{h}_e\| \leq \epsilon_1\right\} \tag{11}$$

where $\mathbf{h}_e$ represents the actual channel vector of the eavesdropper, $\hat{\mathbf{h}}_e$ denotes the estimated channel vector of the eavesdropper, and $\Delta\mathbf{h}_e$ describes the corresponding channel estimation error. The bound error region of the channel estimation is denoted as ϵ_1, which is a constant and can be changed according to the actual requirements.

B. Problem Formulation

The signal is transmitted from the GEO satellite and the receive signal of the ith legitimate user in group k is given by

$$y_i^{\mathrm{geo}} = \mathbf{h}_i^H\mathbf{w}_k s_k^{\mathrm{geo}} + \sum_{\ell \neq k}\mathbf{h}_i^H\mathbf{w}_\ell s_\ell^{\mathrm{geo}} + n_i^{\mathrm{geo}},\ i \in \mathcal{U}_k \tag{12}$$

and similarly, the received signal of the eavesdropper intending to the intercept group k is denoted as

$$y_{e,k}^{\mathrm{geo}} = \mathbf{h}_e^H\mathbf{w}_k s_k^{\mathrm{geo}} + \sum_{\ell \neq k}\mathbf{h}_e^H\mathbf{w}_\ell s_\ell^{\mathrm{geo}} + n_e^{\mathrm{geo}} \tag{13}$$

where $\mathbf{w}_k \in \mathbb{C}^{N_t \times 1}$ and s_k^{geo} are the beamforming vector and transmit signal for group k, respectively. Specifically, the transmit signal satisfies $\mathbb{E}\{s_k^{\mathrm{geo}}\} = 0$ and $\mathbb{E}\{|s_k^{\mathrm{geo}}|^2\} = 1 \forall k$. $n_i^{\mathrm{geo}} \sim \mathcal{CN}(0, \sigma_g^2)$ and $n_e^{\mathrm{geo}} \sim \mathcal{CN}(0, \sigma_g^2)$ are denoted as the zero-mean additive white Gaussian of the legitimate user and the eavesdropper, respectively. The noise variance is given by $\sigma_g^2 = \kappa B^{\mathrm{geo}} T^{\mathrm{geo}}$, where κ, B^{geo} and T^{geo} denote the Boltzmann's constant, bandwidth, and noise temperature at the receiving end of GEO satellite, respectively [17]. According to the signal expression in (12), the signal-to-interference-plus-noise ratio (SINR) of the ith user for the group k is indicated as

$$\mathrm{SINR}_{i,k}^{\mathrm{geo}} \triangleq \frac{\left|\mathbf{w}_k^H\mathbf{h}_i\right|^2}{\sum_{\ell \neq k}\left|\mathbf{w}_\ell^H\mathbf{h}_i\right|^2 + \sigma_g^2}, \quad \forall k, \ell \in \mathcal{K},\ i \in \mathcal{U}_k \tag{14}$$

and the SINR of eavesdropper, which wiretaps the signal of the kth group is represented as

$$\mathrm{SINR}_{e,k}^{\mathrm{geo}} \triangleq \frac{\left|\mathbf{w}_k^H\mathbf{h}_e\right|^2}{\sum_{\ell \neq k}\left|\mathbf{w}_\ell^H\mathbf{h}_e\right|^2 + \sigma_g^2}, \quad \forall k, \ell \in \mathcal{K}. \tag{15}$$

Then, we use the rate formula to find the rates of the users and the eavesdroppers. Specifically, it is assumed that the achievable data rate of the ith user in group k is expressed as $R_{i,k}^{\mathrm{geo}}$, given by

$$R_{i,k}^{\mathrm{geo}} \triangleq \log_2\left(1 + \mathrm{SINR}_{i,k}^{\mathrm{geo}}\right). \tag{16}$$

Then, the rate for group k be the minimum rate of users in this group k is given by

$$R_k^{\mathrm{geo}} \triangleq \min_{i \in \mathcal{U}_k} R_{i,k}^{\mathrm{geo}} \tag{17}$$

and the rate of the eavesdropper wiretapping group k is calculated as

$$R_{e,k}^{\mathrm{geo}} \triangleq \log_2\left(1 + \mathrm{SINR}_{e,k}^{\mathrm{geo}}\right). \tag{18}$$

Considering all user groups and user channel phase uncertainty, we further introduce the worst-case sum secrecy

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rate (WCSSR) as the design objective [35], denoted as

$$R_{\mathrm{wc}}^{\mathrm{geo}} \triangleq \sum_{k=1}^{K} \min_{\|\Delta\mathbf{h}_e\|\le\epsilon_1} \left[\tilde{R}_k^{\mathrm{geo}} - R_{e,k}^{\mathrm{geo}}\right]^+ = \sum_{k=1}^{K} \left[\tilde{R}_k^{\mathrm{geo}} - \max_{\|\Delta\mathbf{h}_e\|\le\epsilon_1} R_{e,k}^{\mathrm{geo}}\right]^+ \tag{19}$$

where $\tilde{R}_k^{\mathrm{geo}} \triangleq \min_{i\in\mathcal{U}_k} \tilde{R}_{i,k}^{\mathrm{geo}}$ and $\tilde{R}_{i,k}^{\mathrm{geo}} \triangleq \mathbb{E}\{R_{i,k}^{\mathrm{geo}}\} = \mathbb{E}\{\log_2(1+\mathrm{SINR}_{i,k}^{\mathrm{geo}})\}$. It can be seen that we consider an approach that employs the expectation to handle the user channel phase uncertainty problem and the data rate function in (16) is averaged with respect to the channel phase error q_i in (7).

Our objective is to maximize the WCSSR in which the CSI of both legitimate users and eavesdroppers are imperfect and all legitimate users satisfy the transmit power constraints. Specifically, the optimization problem can be formulated as

$$\mathcal{R}_1: \max_{\{\mathbf{w}_k\}_{k=1}^K} \quad \sum_{k=1}^{K} \left[\tilde{R}_k^{\mathrm{geo}} - \max_{\|\Delta\mathbf{h}_e\|\le\epsilon_1} R_{e,k}^{\mathrm{geo}}\right]^+ \tag{20a}$$

$$\text{s.t.} \quad \sum_{k=1}^{K} \|\mathbf{w}_k\|_2^2 \le P \tag{20b}$$

where P denotes the total transmit power threshold. Note that a feasible solution $\mathbf{w}_k = \mathbf{0}$ will cause the value of $R_{\mathrm{wc}}^{\mathrm{geo}}$ to be zero, so all feasible beamformers that make the secrecy rate negative are not optimal for the problem (20). Hence, we can omit $[\cdot]^+$ without losing any optimality [18], and problem $\mathcal{R}_1$ is equivalent to

$$\mathcal{R}_2: \max_{\{\mathbf{w}_k\}_{k=1}^K} \quad \sum_{k=1}^{K} \left(\tilde{R}_k^{\mathrm{geo}} - \max_{\|\Delta\mathbf{h}_e\|\le\epsilon_1} R_{e,k}^{\mathrm{geo}}\right) \tag{21a}$$

$$\text{s.t.} \quad \sum_{k=1}^{K} \|\mathbf{w}_k\|_2^2 \le P. \tag{21b}$$

Obviously, we can note that the constrained optimization problem $\mathcal{R}_2$ is nonconvex and hard to handle. Along with necessary transformations, practical algorithm design for this problem is described in the following subsection.

C. Algorithm Design

As can be seen, solving the problem (21) directly is computationally challenging. Therefore, we propose an iterative scheme that utilizes some methods to convert the problem (21) into a more accessible form to handle.

We first introduce auxiliary variables $\{\beta_{1,k}\}_{k=1}^K$ and $\{\beta_{2,k}\}_{k=1}^K$ into problem $\mathcal{R}_2$ to handle the operation of minimization and maximization in the definition of the WCSSR. Therefore, the problem $\mathcal{R}_2$ can be equivalently converted to

$$\mathcal{R}_3: \max_{\substack{\{\mathbf{w}_k\}_{k=1}^K,\\ \{\beta_{1,k},\beta_{2,k}\}_{k=1}^K}} \quad \sum_{k=1}^{K} (\beta_{1,k} - \beta_{2,k}) \tag{22a}$$

$$\text{s.t.} \quad R_{i,k}^{\mathrm{geo}} \ge \beta_{1,k},\ \forall k,\ell\in\mathcal{K},\ i\in\mathcal{U}_k \tag{22b}$$

$$R_{e,k}^{\mathrm{geo}} \le \beta_{2,k},\ \forall k,\ell\in\mathcal{K} \tag{22c}$$

$$\sum_{k=1}^{K} \|\mathbf{w}_k\|_2^2 \le P. \tag{22d}$$

One of the challenges in the problem (22) is that there is no closed-form expression for the ergodic rate function $\tilde{R}_{i,k}^{\mathrm{geo}}$. The Monte Carlo method is the traditional way to settle this problem. Nevertheless, this method is not practical due to the high computational complexity and large memory requirements. Therefore, an approximation form of the average rate with less complexity is utilized to handle this challenge [35], [36], which is denoted as

$$\tilde{R}_{i,k}^{\mathrm{geo}} = \mathbb{E}\left\{\log_2\left(1+\frac{\left|\mathbf{w}_k^H\mathbf{h}_i\right|^2}{\sum_{\ell\ne k}\left|\mathbf{w}_\ell^H\mathbf{h}_i\right|^2+\sigma_g^2}\right)\right\} \approx \log_2\left(1+\frac{\mathbb{E}\left\{\left|\mathbf{w}_k^H\mathbf{h}_i\right|^2\right\}}{\mathbb{E}\left\{\sum_{\ell\ne k}\left|\mathbf{w}_\ell^H\mathbf{h}_i\right|^2\right\}+\sigma_g^2}\right) \triangleq \overline{R}_{i,k}^{\mathrm{geo}}. \tag{23}$$

Note that the approximation in (23) has been proved to be tight through theoretical analysis and numerical results [35]. Next, $\overline{R}_{i,k}$ is adopted to replace $\tilde{R}_{i,k}$ in the following optimization procedure.

Even after employing the approximate rate $\overline{R}_{i,k}$, it is still tricky to handle the problem (22). With this regard, we invoke an efficient approximation technique, i.e., semidefinite relaxation (SDR). Then, the optimization variables $\{\mathbf{w}_k\}_{k=1}^K$ can be converted into $\{\mathbf{W}_k \triangleq \mathbf{w}_k\mathbf{w}_k^H\}_{k=1}^K$. After relaxing the rank constraint, the equivalent formulation of the problem (22) is formulated as

$$\mathcal{R}_4: \max_{\substack{\{\mathbf{W}_k\}_{k=1}^K,\\ \{\beta_{1,k},\beta_{2,k}\}_{k=1}^K}} \quad \sum_{k=1}^{K}(\beta_{1,k}-\beta_{2,k}) \tag{24a}$$

$$\text{s.t.} \quad \log_2\left(1+\frac{\mathrm{Tr}(\mathbf{X}_i'\mathbf{W}_k)}{\sum_{\ell\ne k}\mathrm{Tr}(\mathbf{X}_i'\mathbf{W}_\ell)+\sigma_g^2}\right) \ge \beta_{1,k},\ \forall k,\ell\in\mathcal{K},\ i\in\mathcal{U}_k \tag{24b}$$

$$\log_2\left(1+\frac{\mathbf{h}_e^H\mathbf{W}_k\mathbf{h}_e}{\sum_{\ell\ne k}\mathbf{h}_e^H\mathbf{W}_\ell\mathbf{h}_e+\sigma_g^2}\right) \le \beta_{2,k},\ \forall k,\ell\in\mathcal{K} \tag{24c}$$

$$\sum_{k=1}^{K}\mathrm{Tr}(\mathbf{W}_k) \le P \tag{24d}$$

$$\mathbf{W}_k \succeq \mathbf{0},\ \forall k\in\mathcal{K} \tag{24e}$$

where the transmitting power of each group is given by $\|\mathbf{w}_k\|_2^2 = \mathrm{Tr}(\mathbf{W}_k)$.

As can be noticed, (24b) and (24c) are still nonconvex constraints. Through some mathematical transformations [37], (24b) can be transformed into the

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following form:

$$\begin{aligned}
&\log_2\left(1+\frac{\mathrm{Tr}(\mathbf{X}_i'\mathbf{W}_k)}{\sum_{\ell\neq k}\mathrm{Tr}(\mathbf{X}_i'\mathbf{W}_\ell)+\sigma_g^2}\right)\\
&=\log_2\left(\frac{\sum_{k=1}^{K}\mathrm{Tr}(\mathbf{X}_i'\mathbf{W}_k)+\sigma_g^2}{\sum_{\ell\neq k}\mathrm{Tr}(\mathbf{X}_i'\mathbf{W}_\ell)+\sigma_g^2}\right)\\
&=\log_2\left(\mathrm{Tr}\left(\mathbf{X}_i'\sum_{k=1}^{K}\mathbf{W}_k\right)+\sigma_g^2\right)\\
&\quad-\log_2\left(\mathrm{Tr}\left(\mathbf{X}_i'\sum_{\ell\neq k}\mathbf{W}_\ell\right)+\sigma_g^2\right)\\
&\geq \beta_{1,k}\forall k,\ell\in\mathcal{K},\ i\in\mathcal{U}_k. \qquad (25)
\end{aligned}$$

To make the formula more concise, we convert the aforementioned equation as

$$f_k(\mathbf{W})-g_k(\mathbf{W})\geq\beta_1,\ \forall k,\ell\in\mathcal{K},\ i\in\mathcal{U}_k \qquad (26)$$

where

$$\mathbf{W}\triangleq\{\mathbf{W}_1,\ldots,\mathbf{W}_K\} \qquad (27)$$

$$f_k(\mathbf{W})=\log_2\left(\mathrm{Tr}\left(\mathbf{X}_i'\sum_{k=1}^{K}\mathbf{W}_k\right)+\sigma_g^2\right) \qquad (28)$$

$$g_k(\mathbf{W})=\log_2\left(\mathrm{Tr}\left(\mathbf{X}_i'\sum_{\ell\neq k}\mathbf{W}_\ell\right)+\sigma_g^2\right). \qquad (29)$$

It can be observed that formula (26) is nonconvexity since the $f_k(\mathbf{W})$ and $g_k(\mathbf{W})$ are all concave over $\mathbf{W}$ and convex function subtraction does not guarantee the convexity of the formula. To tackle this problem, we employ a powerful heuristic method, which is called concave–convex procedure (CCP). The basic idea lies in replacing $g_k(\mathbf{W})$ with their first-order Taylor expansion $\triangle g_k(\mathbf{W})$ at the current iteration first, then address this convex optimization problem with an initial feasible point, and continue to the next iteration until convergence [38]. We can use this approach directly because it has been applied and verified in many literatures [39], [40], [41], [42]. Therefore, the constraints in (26) can be recast as

$$f_k(\mathbf{W})-\triangle g_k^{(\lambda)}(\mathbf{W})\geq\beta_1,\ \forall k,\ell\in\mathcal{K},\ i\in\mathcal{U}_k \qquad (30)$$

where

$$\begin{aligned}
\triangle g_k^{(\lambda)}(\mathbf{W})&=g_k\left(\mathbf{W}^{(\lambda)}\right)\\
&\quad+\sum_{a\neq k}\mathrm{Tr}\left\{\left(\frac{\partial g_k\left(\mathbf{W}^{(\lambda)}\right)}{\partial\mathbf{W}_a}\right)^T\left(\mathbf{W}_a-\mathbf{W}_a^{(\lambda)}\right)\right\}. \qquad (31)
\end{aligned}$$

The set of variables $\{\mathbf{W}_1^{(\lambda)},\ldots,\mathbf{W}_K^{(\lambda)}\}$ is denoted as $\mathbf{W}^{(\lambda)}$, and the iteration index of our proposed scheme is represented as λ. Moreover, the derivatives in (31) are calculated as

$$\frac{\partial g_k\left(\mathbf{W}^{(\lambda)}\right)}{\partial\mathbf{W}_a}=\frac{\mathbf{X}_i'^T}{\left(\mathrm{Tr}\left(\mathbf{X}_i'\sum_{\ell\neq k}\mathbf{W}_\ell^{(\lambda)}\right)+\sigma_g^2\right)\ln 2}. \qquad (32)$$

Next, we handle the constraint (24c). Because of the different expressions of eavesdroppers channel uncertainty, the same optimization method of users channel cannot be applied to the problem of nonconvex constraint (24c). With this regard, we need to use the following lemma to optimize the problem further.

LEMMA 1 (S-PROCEDURE) Define the function

$$y_k(\mathbf{x})=\mathbf{x}^H\mathbf{Y}_k\mathbf{x}+2\mathrm{Re}\{\mathbf{z}_k^H\mathbf{x}\}+d_m,\ k\in\{1,2\} \qquad (33)$$

where $\mathbf{Y}_k\in\mathbb{C}^{N_t\times N_t}$, $\mathbf{z}_k\in\mathbb{C}^{N_t\times 1}$, and $d_m\in\mathbb{R}$. Then, the implication $y_1(\mathbf{x})\leq 0\Rightarrow y_2(\mathbf{x})\leq 0$ holds if and only if there exists μ such that

$$\mu\begin{bmatrix}\mathbf{Y}_1 & \mathbf{z}_1\\ \mathbf{z}_1^H & d_1\end{bmatrix}-\begin{bmatrix}\mathbf{Y}_2 & \mathbf{z}_2\\ \mathbf{z}_2^H & d_2\end{bmatrix}\succeq\mathbf{0} \qquad (34)$$

provided that there exists a point $\widehat{\mathbf{x}}$ such that $y_k(\widehat{\mathbf{x}})<0$.

By using the Taylor series expansion and S-Procedure of *Lemma 1*, we can obtain the proposition as follows.

PROPOSITION 1 By introducing the auxiliary variables $\{\beta_{3,k}\}_{k=1}^K$ and $\{\beta_{4,k}\}_{k=1}^K$, the approximation of constraint (24c) can be reformulated as

$$\beta_{2,k}+\log_2\beta_{3,k}-\log_2\beta_{4,k}^{(\lambda)}-\frac{\beta_{4,k}-\beta_{4,k}^{(\lambda)}}{\beta_{4,k}^{(\lambda)}\ln 2}\geq 0 \qquad (35a)$$

$$\begin{bmatrix}\mu_{1,k}\mathbf{I}_{N_t}+\mathbf{F}_1 & \mathbf{F}_1\hat{\mathbf{h}}_e\\ \hat{\mathbf{h}}_e^H\mathbf{F}_1 & \Gamma_k\end{bmatrix}\succeq\mathbf{0} \qquad (35b)$$

$$\begin{bmatrix}\mu_{2,k}\mathbf{I}_{N_t}-\mathbf{F}_2 & -\mathbf{F}_2\hat{\mathbf{h}}_e\\ -\hat{\mathbf{h}}_e^H\mathbf{F}_2 & \Phi_k\end{bmatrix}\succeq\mathbf{0} \qquad (35c)$$

where $\{\mu_{1,k}\}_{k=1}^K\geq 0$ and $\{\mu_{2,k}\}_{k=1}^K\geq 0$ are slack variables, $\mathbf{F}_1=\sum_{\ell\neq k}(\mathbf{W}_\ell)$, $\Gamma_k=-\mu_{1,k}\epsilon_1^2+\hat{\mathbf{h}}_e^H\mathbf{F}_1\hat{\mathbf{h}}_e-\beta_{3,k}+\sigma_g^2$, $\mathbf{F}_2=\sum_{k=1}^K(\mathbf{W}_k)$, and $\Phi_k=-\mu_{2,k}\epsilon_1^2-\hat{\mathbf{h}}_e^H\mathbf{F}_2\hat{\mathbf{h}}_e+\beta_{4,k}-\sigma_g^2$.

Proof: See Appendix V-A. ■

Combining the aforementioned transformations into the problem (24), we obtain a series of optimization formulated as

$$\mathcal{R}_5:\ \max_{\{\mathbf{W}_k,\mu_{1,k},\mu_{2,k},\beta_{1,k},\beta_{2,k},\beta_{3,k},\beta_{4,k}\}_{k=1}^K}\ \sum_{k=1}^{K}(\beta_{1,k}-\beta_{2,k}) \qquad (36a)$$

$$\text{s.t. (30), (35a), (35b), (35c), (24d), (24e)} \qquad (36b)$$

$$\beta_{3,k}\geq 0,\ \beta_{4,k}\geq 0,\ \mu_{1,k}\geq 0,\ \mu_{2,k}\geq 0,\ \forall k\in\mathcal{K}. \qquad (36c)$$

which can be solved efficiently. It is worth pointing out that if the rank of results $\{\mathbf{W}_k\}_{k=1}^K$ are all equal to one, then the optimal beamforming solution $\{\mathbf{w}_k\}_{k=1}^K$ of the original problem (20) can be obtained by eigenvalue decomposition. Otherwise, we can acquire the suboptimal solution of original problem by utilizing the randomization technique [37]. As a result, the whole procedure of problem $\mathcal{R}_1$ is shown in Algorithm 1.

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Algorithm 1: Robust Secure Transmission Design for GEO Satellite Systems.

Input: Initial feasible beamformer correlation matrices $\{\mathbf{W}_k^{(0)}\}_{k=1}^K$.

Output: The optimal beamformer vectors $\{\mathbf{w}_k^*\}_{k=1}^K$.

1: Initialization: iteration threshold ς_1 and index $\lambda = 0$.
2: **repeat**
3: Calculate the convex problem (36).
4: Obtain the solution $\{\mathbf{W}_k^{\text{temp}}\}_{k=1}^K$ and $\{\beta_{4,k}^{\text{temp}}\}_{k=1}^K$.
5: Update $\{\mathbf{W}_k^{(\lambda+1)}\}_{k=1}^K \leftarrow \{\mathbf{W}_k^{\text{temp}}\}_{k=1}^K$ and $\{\beta_{4,k}^{(\lambda+1)}\}_{k=1}^K \leftarrow \{\beta_{4,k}^{\text{temp}}\}_{k=1}^K$. Set $\lambda \leftarrow \lambda + 1$.
6: **until** $|R_{\text{wc}}^{\text{geo},(\lambda)} - R_{\text{wc}}^{\text{geo},(\lambda-1)}| \le \varsigma_1$.
7: Return
8: Set $\{\mathbf{W}_k^*\}_{k=1}^K \leftarrow \{\mathbf{W}_k^{(\lambda)}\}_{k=1}^K$.
9: According to the rank of $\{\mathbf{W}_k^*\}_{k=1}^K$, judge whether to use eigenvalue decomposition or randomization to get the final solution.

D. Complexity and Convergence Analysis

We can note that problem (36) is a convex semidefinite program (SDP) problem that can be solved iteratively. On this basis, the computational complexity of Algorithm 1 is studied by employing the interior point method (IPM). Assuming that the number of iterations is N^{geo}, and according to the complexity analysis of the IPM described in [43], Remark 3.13], the computational complexity for Algorithm 1 is $\mathcal{O}(N^{\text{geo}}KN_{\text{t}}^{7/2} + 4N^{\text{geo}}K^2N_{\text{t}}^{3/2} + 8N^{\text{geo}}K^3N_{\text{t}}^{1/2})\log(1/\varsigma_1)$, where ς_1 is the predefined accuracy [43].

Furthermore, we research the convergence of the secure transmission algorithm of GEO satellite. Since the problem (36) is a typical convex problem, the solution $\{\mathbf{W_k}\}_{k=1}^K$ and $\{\beta_{4,k}\}_{k=1}^K$ will be updated at each iteration and it is always a feasible point of the next iteration. As a result, the WC-SSR obtained in problem (36) will be monotonically non-decreasing through the proposed iterative algorithm [12], [44].

III. ROBUST SECURE TRANSMISSION FOR THE LEO SATELLITE SYSTEM

Consistent with the GEO satellite systems, the LEO satellite systems also demand considering the robust secure transmission problem. For LEO satellites, because of their relatively less delay, smaller path loss, and the relatively small number of users served by a single satellite, each spot beam usually serves only one user. Therefore, in this section, we consider the unicast secure transmission for LEO satellite communication systems.

In GEO satellite systems, reflector antennas are usually utilized to realize multibeam transmission, and channel phase uncertainty is mainly considered in channel modeling. However, LEO satellite systems generally adopt the uniform planar array (UPA). Since the LEO satellites have the characteristics of high-speed mobility, the transmitter cannot obtain the accurate angle information of users, and the channel response vectors have the problem of angle uncertainty, which is different from GEO satellite channel modeling. Based on this, we consider the secure transmission designs of the LEO satellite with channel angular uncertainties in this section.

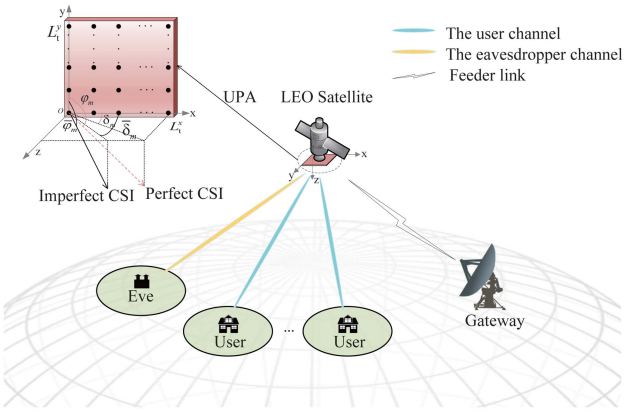


Fig. 2. Model of a downlink LEO satellite communication system.

A. Channel Model

In this section, we consider a downlink LEO satellite secure transmission system consisted of a satellite, M legitimate users, and one eavesdropper. The LEO satellite adopts a large-scale UPA composed of L_{t} antennas. We denote $L_{\text{t}} \triangleq L_{\text{t}}^x L_{\text{t}}^y$, where L_{t}^x and L_{t}^y are the number of elements in the x- and y-axes, respectively. In addition, each user and eavesdropper taken into account are single antenna in this article, and the eavesdropper adopts passive mode to wiretap the channels between satellite and users. The system model of the LEO satellite is represented in Fig. 2.

1) *Channel Propagation Model:* It is assumed that the channel links connecting the LEO satellite and different legitimate users are uncorrelated because different users are usually separated in space [45]. To capture the Doppler effect caused by the LEO satellite movement, we adopt a ray-tracing-based model, which is different to that for the GEO case, to express the complex baseband downlink spatial channel response of the LEO satellite and the mth user at instant t and frequency f as [46]

$$\hat{\mathbf{g}}_m = \exp\left\{\jmath 2\pi\left[t\nu_m^{\text{sat}} - f\tau_m^{\min}\right]\right\} \cdot g_m^{\text{dl}}(t,f) \cdot \hat{\mathbf{v}}_m \in \mathbb{C}^{L_{\text{t}}\times 1} \tag{37}$$

where ν_m^{sat} is Doppler shifts caused by the LEO satellite movement, and $\tau_m^{\min} = \min_k\{\tau_{m,k}\}$ represents the minimum propagation delay over all paths of the mth user. The downlink channel gain $g_m^{\text{dl}}(t,f)$ of the mth user is expressed as

$$g_m^{\text{dl}}(t,f) = \sum_{k=0}^{N_m-1} g_{m,k}^{\text{dl}} \exp\left\{\jmath 2\pi\left[t\nu_{m,k}^{\text{ut}} - f\tau_{m,k}^{\text{ut}}\right]\right\} \tag{38}$$

where N_m is the number of transmission paths of the mth user, $g_{m,k}^{\text{dl}}$ denotes the complex-valued gain with the path k of the mth user. $\nu_{m,k}^{\text{ut}}$ and $\tau_{m,k}^{\text{ut}}$ represent the Doppler shifts and propagation delay, which caused by the user movement,

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respectively, and $\tau_{m,k}^{\mathrm{ut}}$ is defined as $\tau_{m,k}^{\mathrm{ut}} = \tau_{m,k} - \tau_m^{\min}$. In this article, it is assumed that the delay and Doppler have been atoned for legitimate users and eavesdropper [46]. Hence, the time and frequency of satellite, legitimate users, and eavesdropper are perfectly synchronized. It is worth noting that LEO satellite systems are often moved under line-of-sight (LOS) propagation. In addition, in LOS satellite communication systems, the Rician channel model is widely employed. Based on this, channel gain $g_{m,k}^{\mathrm{dl}}$ is assumed to satisfy Rician fading distribution with the Rician factor κ_m and $\mathbb{E}\{|g_{m,k}^{\mathrm{dl}}(t,f)|^2\} = \gamma_m$ [46]. Moreover, the UPA response vector $\hat{\mathbf{v}}_m$ is given by [47]

$$\hat{\mathbf{v}}_m \triangleq \hat{\mathbf{v}}_x(\delta_m, \varphi_m) \otimes \hat{\mathbf{v}}_y(\delta_m) \in \mathbb{C}^{L_{\mathrm{t}} \times 1} \tag{39}$$

where δ_m denotes the angles of the mth user relative to the x-axis and φ_m describes the angles of the mth user with respect to y-axis. In addition, $\hat{\mathbf{v}}_x(\delta_m, \varphi_m)$ is the UPA response vector of the angle related to the x-axis expressed as

$$\hat{\mathbf{v}}_x(\delta_m, \varphi_m) = \frac{1}{\sqrt{L_{\mathrm{t}}^x}} \left[1, \exp\{-\jmath\pi \sin(\delta_m)\cos(\varphi_m)\} \right. \\ \left. \ldots, \exp\left\{-\jmath\pi\left(L_{\mathrm{t}}^x - 1\right)\sin(\delta_m)\cos(\varphi_m)\right\}\right]^T. \tag{40}$$

Similarly, $\hat{\mathbf{v}}_y(\delta_m)$, which denotes the UPA response vector of angle related to the y-axis, is expressed as

$$\hat{\mathbf{v}}_y(\delta_m) = \frac{1}{\sqrt{L_{\mathrm{t}}^y}} \left[1, \exp\{-\jmath\pi \cos(\delta_m)\} \right. \\ \left. \ldots, \exp\left\{-\jmath\pi\left(L_{\mathrm{t}}^y - 1\right)\cos(\delta_m)\right\}\right]^T. \tag{41}$$

2) *Channel Angle Uncertainty Model of Legitimate User:* In the practical LEO satellite downlink communication scenarios, the transmitter often only knows the imperfect angular information of each user because of the satellites high-speed mobility, making the radiated signals from the UPA unable to accurately point to the served users. This may incur severe performance degradation in the transmission efficiency of the satellite system. Therefore, we focus on the actual angle information received on the satellite side, which can be modeled as

$$\delta_m = \hat{\delta}_m + \bar{\delta}_m + \tilde{\delta} \tag{42}$$

$$\varphi_m = \hat{\varphi}_m + \bar{\varphi}_m + \tilde{\varphi} \tag{43}$$

where $\hat{\delta}_m$ and $\hat{\varphi}_m$ are the angle estimates of the mth user in the x- and y-axes directions, respectively. However, they are always inaccurate. With this regard, we introduce the angular deviation caused by angle estimation error, expressed as $\bar{\delta}_m$ and $\bar{\delta}_m$. They can be modeled as random variables with a particular distribution (e.g., uniform distribution), respectively. In addition, $\tilde{\delta}$ and $\tilde{\varphi}$ represent the common angel deviation for all users caused by the attitude jitter of the satellite [48], [49]. Note that $\tilde{\delta}$ and $\tilde{\varphi}$ can be omitted because the estimation error can be controlled within 10^{-4} by the current attitude estimation technologies for satellite systems so that the actual effect of $\widetilde{\delta}$ and $\widetilde{\varphi}$ is very small. Based on the aforementioned angle model and formula (39), we can get the actual UPA response vector $\hat{\mathbf{v}}_m$, which is represented as

$$\mathbf{v}_m \triangleq \mathbf{v}_x\left(\hat{\delta}_m + \bar{\delta}_m, \hat{\varphi}_m + \bar{\varphi}_m\right) \otimes \mathbf{v}_y\left(\hat{\delta}_m + \bar{\delta}_m\right). \tag{44}$$

According to the (37) and (44), the actual channel response of LEO satellite can be modeled as

$$\mathbf{g}_m = \exp\left\{\jmath 2\pi\left[t\nu_m^{\mathrm{sat}} - f\tau_m^{\min}\right]\right\} \cdot g_m^{\mathrm{dl}}(t,f) \cdot \mathbf{v}_m. \tag{45}$$

In the following sections, we assume that the LEO satellite adopt the imperfect CSI of each user and the angular deviations follow a uniform distribution.

3) *Channel Uncertainty Model of Eavesdropper:* Referring to the eavesdropper channel model of the GEO satellite, the specific information of CSI is uncertain, therefore, we assume that the satellite only knows the channel uncertainty region of the eavesdropper, which is modeled as

$$\Omega_e^{\mathrm{leo}} = \left\{\mathbf{g}_e \mid \mathbf{g}_e = \hat{\mathbf{g}}_e + \Delta\mathbf{g}_e = g_e^{\mathrm{dl}} \cdot \hat{\mathbf{v}}_e^H \right. \\ \left. + g_e^{\mathrm{dl}} \cdot \Delta\mathbf{v}_e^H, \|\Delta\mathbf{v}_e\| \le \epsilon_2\right\} \tag{46}$$

where $\mathbf{g}_e$ represents the channel estimate, $\Delta\mathbf{g}_e \triangleq g_e^{\mathrm{dl}} \cdot \Delta\mathbf{v}_e^H$ is the corresponding channel estimation error, and ϵ_2 represents the bound of the uncertainty region.

B. Problem Formulation

Denote $\mathcal{M} = \{1, \ldots, M\}$ as the user index set. The received signal of the mth user is expressed as

$$y_m = \mathbf{g}_m^H \mathbf{b}_m s_m^{\mathrm{leo}} + \sum_{n \ne m} \mathbf{g}_m^H \mathbf{b}_n s_n^{\mathrm{leo}} + n_m^{\mathrm{leo}}, \ \forall m, n \in \mathcal{M} \tag{47}$$

and the received signal for the eavesdropper is

$$y_e = \sum_{m=1}^{M} \mathbf{g}_e^H \mathbf{b}_m s_m^{\mathrm{leo}} + n_e^{\mathrm{leo}}, \ \forall m \in \mathcal{M} \tag{48}$$

where $\mathbf{b}_m \in \mathbb{C}^{L_{\mathrm{t}} \times 1}$ describes the precoding vector for the user m. s_m^{leo} is the LEO satellite transmit signal for the mth user with $\mathbb{E}\{s_m^{\mathrm{leo}}\} = 0$ and $\mathbb{E}\{|s_m^{\mathrm{leo}}|^2\} = 1 \forall m$. n_m^{leo} and n_e^{leo} are the complex-valued additive Gaussian white noise, which distribute as $n_m^{\mathrm{leo}} \sim \mathcal{CN}(0, \sigma_m^2)$ and $n_e^{\mathrm{leo}} \sim \mathcal{CN}(0, \sigma_e^2)$, respectively. In this article, the noise variances are assumed to be equal, i.e., $\sigma_m^2 = \sigma_e^2 = \sigma_l^2$, and satisfy $\sigma_l^2 = \kappa B^{\mathrm{leo}} T^{\mathrm{leo}}$, where B^{leo} represents the bandwidth of LEO satellite systems and T^{leo} describes the noise temperature [50].

Based on (47), the SINR of the mth user for LEO satellite systems is given by

$$\mathrm{SINR}_m^{\mathrm{leo}} \triangleq \frac{\left|\mathbf{b}_m^H \mathbf{g}_m\right|^2}{\sum_{n \ne m}\left|\mathbf{b}_n^H \mathbf{g}_m\right|^2 + \sigma_l^2}, \quad \forall m, n \in \mathcal{M}. \tag{49}$$

According to the aforementioned formula, we can obtain the rate of the mth user as

$$R_m^{\mathrm{leo}} \triangleq \mathbb{E}\left\{\log_2\left(1 + \mathrm{SINR}_m^{\mathrm{leo}}\right)\right\} \\ = \mathbb{E}\left\{\log_2\left(1 + \frac{\left|\mathbf{b}_m^H \mathbf{g}_m\right|^2}{\sum_{n \ne m}\left|\mathbf{b}_n^H \mathbf{g}_m\right|^2 + \sigma_l^2}\right)\right\}. \tag{50}$$

On the other hand, we assume that in the worst case, the eavesdropper can eliminate all users interference signals

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except the interested [51], then we can get the rate of the eavesdropper as follows:

$$R_{e,m}^{\text{leo}} \triangleq \mathbb{E}\left\{\log_2\left(1+\frac{1}{\sigma_l^2}\left|\mathbf{b}_m^H\mathbf{g}_e\right|^2\right)\right\}. \tag{51}$$

Therefore, the minimum secrecy rate of the mth user is denoted as

$$\begin{aligned} R_{\text{wc},m}^{\text{leo}} &\triangleq \min_{\|\Delta\mathbf{v}_e\|\le\epsilon_2}\left[R_m^{\text{leo}}-R_{e,m}^{\text{leo}}\right]^+ \\ &= \left[R_m^{\text{leo}}-\max_{\|\Delta\mathbf{v}_e\|\le\epsilon_2}R_{e,m}^{\text{leo}}\right]^+. \end{aligned} \tag{52}$$

Note that similar to the GEO satellite optimization problem, $[\cdot]^+$ can be omitted without loss of optimality.

In this section, we consider the robust secure transmission in the LEO satellite systems, and our objective is to maximize the WCSSR with the transmit power constraints. This constrained optimization problem can be formulated as

$$\mathcal{F}_1: \max_{\{\mathbf{b}_m\}_{m=1}^M} \quad R_{\text{wc}}^{\text{leo}} = \sum_{m=1}^M\left[R_m^{\text{leo}}-\max_{\|\Delta\mathbf{v}_e\|\le\epsilon_2}R_{e,m}^{\text{leo}}\right] \tag{53a}$$

$$\text{s.t.} \quad \sum_{m=1}^M\|\mathbf{b}_m\|_2^2\le P. \tag{53b}$$

The problem $\mathcal{F}_1$ is challenging to deal with directly for the reason that the average rates do not admit explicit expressions. Besides, the imperfect CSI further increases the computational complexity of the constrained optimization problem.

C. Algorithm Design

It can be noted that solving the problem directly is computationally challenging, thus we propose an iterative algorithm, which is based on the S-Procedure, CCP, and Taylor series expansion to handle the problem $\mathcal{F}_1$ efficiently.

First of all, we introduce the auxiliary variables $\{\log_2\alpha_m\}_{m=1}^M$, where $\log_2\alpha_m$ is the maximum rate of eavesdropping rate of the mth user. Then, problem (53) can be equivalently rewritten as

$$\mathcal{F}_2: \max_{\{\mathbf{b}_m,\log_2\alpha_m\}_{m=1}^M} \quad \sum_{m=1}^M\left[R_m^{\text{leo}}-\log_2\alpha_m\right] \tag{54a}$$

$$\text{s.t.} \quad R_{e,m}^{\text{leo}}\le\log_2\alpha_m,\ \forall m\in\mathcal{M} \tag{54b}$$

$$\sum_{m=1}^M\|\mathbf{b}_m\|_2^2\le P. \tag{54c}$$

Similar to the previous optimization for GEO systems, we employ an approximation for the average rates and introduce new matrices $\mathbf{B}_m=\mathbf{b}_m\mathbf{b}_m^H$ to get [50]

$$\begin{aligned} R_m^{\text{leo}} &\approx \overline{R}_m^{\text{leo}} \triangleq \log_2\left(1+\frac{\mathbb{E}\left\{\left|\mathbf{b}_m^H\mathbf{g}_m\right|^2\right\}}{\mathbb{E}\left\{\sum_{n\ne m}\left|\mathbf{b}_n^H\mathbf{g}_m\right|^2\right\}+\sigma_l^2}\right) \\ &= \log_2\left(1+\frac{\gamma_m\text{Tr}(\mathbf{V}_m\mathbf{B}_m)}{\gamma_m\sum_{n\ne m}\text{Tr}(\mathbf{V}_m\mathbf{B}_n)+\sigma_l^2}\right) \\ &= \underbrace{\log_2\left(\gamma_m\text{Tr}\left(\mathbf{V}_m\sum_{m=1}^M\mathbf{B}_m\right)+\sigma_l^2\right)}_{f_m(\mathbf{B})} \\ &\quad -\underbrace{\log_2\left(\gamma_m\text{Tr}\left(\mathbf{V}_m\sum_{n\ne m}\mathbf{B}_n\right)+\sigma_l^2\right)}_{g_m(\mathbf{B})} \end{aligned} \tag{55}$$

and

$$\begin{aligned} R_{e,m}^{\text{leo}} &\approx \overline{R}_{e,m}^{\text{leo}} \triangleq \log_2\left(1+\mathbb{E}\left\{\frac{1}{\sigma_l^2}\left|\mathbf{b}_m^H\mathbf{g}_e\right|^2\right\}\right) \\ &= \log_2\left(1+\frac{\gamma_e}{\sigma_l^2}\mathbf{v}_e^H\mathbf{B}_m\mathbf{v}_e\right) \end{aligned} \tag{56}$$

where $\mathbf{B}\triangleq\{\mathbf{B}_m\}_{m=1}^M$ and $\mathbf{V}_m=\mathbb{E}\{\mathbf{v}_m\mathbf{v}_m^H\}$. The response vector of eavesdropper is represented as $\mathbf{v}_e$ and $\mathbf{v}_e=\hat{\mathbf{v}}_e+\Delta\mathbf{v}_e$. The (k,l)th element of the matrix $\mathbf{V}_m$ is expressed as (57) shown at the bottom of this page, where

$$\begin{aligned} C &= (b-a)\cos(\hat{\varphi}_m)\cos\left(\hat{\delta}_m\right) \\ &\quad -\left(l-k-(b-a)L_{\text{t}}^x\right)\sin\left(\hat{\delta}_m\right) \end{aligned} \tag{58a}$$

$$D = \pi(b-a)\sin(\hat{\varphi}_m) \tag{58b}$$

$$\begin{aligned} Z &= \exp\left\{\jmath\pi\left(l-k-(b-a)L_{\text{t}}^y\right)\cos\left(\hat{\delta}_m\right)\right. \\ &\quad \left.+\jmath\pi(b-a)\cos(\hat{\varphi}_m)\sin\left(\hat{\delta}_m\right)\right\} \end{aligned} \tag{58c}$$

and $a=\lfloor\frac{k}{L_{\text{t}}^y}\rfloor$, $b=\lfloor\frac{l}{L_{\text{t}}^y}\rfloor$. We assume the angle estimation error $\bar{\delta}_m$ and $\bar{\varphi}_m$ both exhibit a uniform distribution, which can be denoted as $\bar{\delta}_m\sim U[\delta_L,\delta_U]$ and $\bar{\varphi}_m\sim U[\varphi_L,\varphi_U]$, and then, we can obtain the $\Delta_\delta=\delta_U-\delta_L$ and $\Delta_\varphi=\varphi_U-\varphi_L$.

It is observed that (55) can be handled by employing the CCP approach, and then, the first-order Taylor series expansion is utilized to process the latter term of the objective function of (54). Hence, we can obtain an equivalent

$$[\mathbf{V}_m]_{k,l}=\mathbb{E}\left\{\left[\mathbf{v}_m\mathbf{v}_m^H\right]_{k,l}\right\}=\begin{cases}\frac{1}{L_{\text{t}}^xL_{\text{t}}^y}, & l=k,a=b\\ \frac{1}{L_{\text{t}}^xL_{\text{t}}^y}\frac{1}{\Delta_\delta}\frac{2\exp\{\jmath\pi(l-k)\cos(\hat{\delta}_m)\}\sin(\pi(l-k)\sin(\hat{\delta}_m)\delta_U)}{\pi(l-k)\sin(\hat{\delta}_m)}, & l\ne k,a=b\\ \frac{1}{L_{\text{t}}^xL_{\text{t}}^y}\frac{4Z}{\Delta_\delta\Delta_\varphi}\frac{\sin(D\varphi_U\sin(\hat{\delta}_m))\sin(\pi C\delta_U)}{\pi CD\sin(\hat{\delta}_m)}, & l\ne k,a\ne b\end{cases} \tag{57}$$

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form of $\mathcal{F}_2$, which is expressed as

$$\mathcal{F}_3: \max_{\substack{\{\mathbf{B}_m\}_{m=1}^M,\\ \{\log_2 \alpha_m\}_{m=1}^M}} \quad \sum_{m=1}^{M}\Big[f_m(\mathbf{B}) - \triangle g_m^{(\tau)}(\mathbf{B}) - \log_2 \alpha_m^{(\tau)} - \frac{\alpha_m - \alpha_m^{(\tau)}}{\alpha_m^{(\tau)} \ln 2}\Big] \tag{59a}$$

$$\text{s.t.} \quad \log_2\left(1 + \frac{\gamma_e}{\sigma_l^2}\mathbf{v}_e^H \mathbf{B}_m \mathbf{v}_e\right) \le \log_2 \alpha_m, \ \forall m \in \mathcal{M} \tag{59b}$$

$$\sum_{m=1}^{M} \mathrm{Tr}(\mathbf{B}_m) \le P \tag{59c}$$

$$\mathbf{B}_m \succeq \mathbf{0}, \ \forall m \in \mathcal{M} \tag{59d}$$

where

$$\triangle g_m^{(\tau)}(\mathbf{B}) = g_m\left(\mathbf{B}^{(\tau)}\right) + \sum_{j \ne m} \mathrm{Tr}\left\{\left(\frac{\partial g_m\left(\mathbf{B}^{(\tau)}\right)}{\partial \mathbf{B}_j}\right)^T \left(\mathbf{B}_j - \mathbf{B}_j^{(\tau)}\right)\right\} \tag{60}$$

$\mathbf{B}^{(\tau)}$ refers to an aggregation of variables $\{\mathbf{B}_1^{(\tau)}, \ldots, \mathbf{B}_M^{(\tau)}\}$, and τ is the iteration index. In addition, the derivatives in (60) are calculated as

$$\frac{\partial g_m\left(\mathbf{B}^{(\tau)}\right)}{\partial \mathbf{B}_j} = \frac{{\mathbf{V}_m}^T}{\left(\gamma_m \mathrm{Tr}\left(\mathbf{V}_m \sum_{n \ne m} \mathbf{B}_n^{(\tau)}\right) + \sigma_l^2\right) \ln 2}. \tag{61}$$

Since the eavesdropper has channel errors, it is challenging to deal with the constraint (59b) directly. With this regard, we need to further optimize it. First, through simple mathematical changes [44], the constraint can be rewritten as

$$\mathbf{v}_e^H \mathbf{B}_m \mathbf{v}_e \le \frac{\sigma_l^2}{\gamma_e}(\alpha_m - 1). \tag{62}$$

Considering the channel uncertainty in (46), we can convert the aforementioned constraint to

$$\begin{cases} \forall \triangle \mathbf{v}_e : \triangle \mathbf{v}_e^H \triangle \mathbf{v}_e - \epsilon_2^2 \le 0 \\ \triangle \mathbf{v}_e^H \mathbf{B}_m \triangle \mathbf{v}_e + 2\mathrm{Re}\left\{\hat{\mathbf{v}}_e^H \mathbf{B}_m \triangle \mathbf{v}_e\right\} + \eta_m \le 0 \end{cases} \tag{63}$$

where $\eta_m = \hat{\mathbf{v}}_e^H \mathbf{B}_m \hat{\mathbf{v}}_e - \frac{\sigma_l^2}{\gamma_e}(\alpha_m - 1)$. Then, by using *Lemma 1*, the constraint (59b) can be converted into the linear matrix inequalities (LMIs), expressed as follows:

$$\begin{bmatrix} s_m \mathbf{I}_{L_t} - \mathbf{B}_m & -\mathbf{B}_m \hat{\mathbf{v}}_e \\ -\hat{\mathbf{v}}_e^H \mathbf{B}_m & \Phi_m \end{bmatrix} \succeq \mathbf{0} \tag{64}$$

where $\{s_m\}_{m=1}^M \ge 0$ is a slack variable and

$$\Phi_m = -s_m \epsilon_2^2 - \hat{\mathbf{v}}_e^H \mathbf{B}_m \hat{\mathbf{v}}_e + \frac{\sigma_l^2}{\gamma_e}(\alpha_m - 1). \tag{65}$$

Algorithm 2: Robust Secure Beamforming Design for LEO Satellite Systems.

Input: Initial feasible beamformer correlation matrices $\{\mathbf{B}_m^{(\tau)}\}_{m=1}^M$, iteration index $\tau = 0$ and threshold ς_2.

Output: The optimal beamformer vectors $\{\mathbf{b}_m^*\}_{m=1}^M$.

1: **while** $|R_{\mathrm{wc}}^{\mathrm{leo},(\tau)} - R_{\mathrm{wc}}^{\mathrm{leo},(\tau-1)}| > \varsigma_2$ **do**
2: Solve the convex problem (66) to obtain the solution $\{\mathbf{B}_m^{\mathrm{temp}}\}_{m=1}^M$ and $\{\alpha_m^{\mathrm{temp}}\}_{m=1}^M$.
3: Set $\tau \leftarrow \tau + 1$.
4: Update $\{\mathbf{B}_m^{(\tau)}\}_{m=1}^M \leftarrow \{\mathbf{B}_m^{(\tau-1)}\}_{m=1}^M$, $\{\alpha_m^{(\tau)}\}_{m=1}^M \leftarrow \{\alpha_m^{(\tau)}\}_{m=1}^M$.
5: **end while**
6: Set $\{\mathbf{B}_m^*\}_{m=1}^M \leftarrow \{\mathbf{B}_m^{(\tau)}\}_{m=1}^M$.
7: Use eigenvalue decomposition for $\{\mathbf{B}_m^*\}_{m=1}^M$ to get the final solution $\{\mathbf{b}_m^*\}_{m=1}^M$.

Thus, according to the aforementioned transformation, the problem $\mathcal{F}_3$ can be transformed into

$$\mathcal{F}_4: \max_{\substack{\{\mathbf{B}_m\}_{m=1}^M,\\ \{\log_2 \alpha_m, s_m\}_{m=1}^M}} \quad R_{\mathrm{wc}}^{\mathrm{leo}} = \sum_{m=1}^{M}\Big[f_m(\mathbf{B}) - \triangle g_m^{(\tau)}(\mathbf{B}) - \log_2 \alpha_m^{(\tau)} - \frac{\alpha_m - \alpha_m^{(\tau)}}{\alpha_m^{(\tau)} \ln 2}\Big] \tag{66a}$$

$$\text{s.t.} \quad (59c), \ (59d), \ (64), \tag{66b}$$

$$s_m \ge 0, \ \forall m \in \mathcal{M}. \tag{66c}$$

With the aforementioned transformation, the original problem $\mathcal{F}_1$ becomes a convex SDP problem, which can be solved through standard mathematical optimization tools. Finally, the robust secure transmission design for LEO satellite systems is summarized as Algorithm 2.

D. Complexity and Convergence Analysis

Note that (66) is a convex SDP problem. The computational complexity of Algorithm 2 is $\mathcal{O}(N^{\mathrm{leo}} M L_{\mathrm{t}}^{7/2} + N^{\mathrm{leo}} M^2 L_{\mathrm{t}}^{3/2} + 8 N^{\mathrm{leo}} M^3 L_{\mathrm{t}}^{1/2}) \log(1/\varsigma_2)$, where N^{leo} denotes the number of iterations, ς_2 represents the given accuracy [43]. In addition, the WCSSR obtained in the problem (66) is monotonically nondecreasing through the proposed iterative algorithm and it can converge to a stationary point [12], [44].

IV. SIMULATION RESULTS

This section presents the numerical analysis to demonstrate the performance of the proposed robust secure transmission methods for GEO and LEO systems. For the GEO satellite systems, the channel phase error variances of different users are assumed to be the same, i.e., $\sigma_i^2 = \sigma_u^2$. The number of beams N_{t} is set to 7, and that of users in group $\mathcal{U}_k$ is set to 5. The specific simulation parameters of GEO satellite systems are shown in Table II . For the LEO satellite systems, the channel is inaccurate and there exists the angle estimation error, so we assume that the angular

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TABLE II
Simulation Parameters for GEO Satellite Systems

Parameter	Value
Orbit altitude d_0	3.6×10^4 km
Boltzmann's constant κ	1.38×10^{-23} J/K
Carrier frequency f	20 GHz
Hexagonal beam length	250 km
Satellite antenna gain G_T^j	38 dBi
Noise bandwidth B^{geo}	50 MHz
Receiver gain to noise temperature $G_{r,i}/T$	15 dB/K
Rain fading variance	1.63 dB
Rain fading mean	-2.6 dB
3-dB angle $\theta_{3\,\text{dB}}$	0.4°

TABLE III
Simulation Parameters for LEO Satellite Systems

Parameter	Value
Earth radius R_e	6378 km
Orbit altitude H	1000 km
Carrier frequency f	20 GHz
Bandwith B^{leo}	50 MHz
Noise temperature T^{leo}	290 K
Number of antennas L_{t}	128
Number of antennas on $x-$ and $y-$axes	8×16
Number of service users	4
Per-Antenna gain G_{sat}, G_{ut}	3 dB, 0 dB
Channel Rician factor $\kappa_k = \kappa$	5 dB
Estimated angles $\hat{\delta}_m$ for M users	$0^\circ < \hat{\delta}_m < 180^\circ$
Estimated angles $\hat{\varphi}_m$ for M users	$0^\circ < \hat{\varphi}_m < 180^\circ$
Angular deviation $\bar{\delta}_m = \bar{\delta}$, $\bar{\varphi}_m = \bar{\varphi}$	$-5^\circ \sim 5^\circ$
Uniform distribution $[\delta_L/\varphi_L, \delta_L/\varphi_U]$	$[-5^\circ, 5^\circ]$

deviations $\bar{\delta}$ and $\bar{\varphi}$ are independently and distributed as uniform distribution. The average channel power is calculated by $\gamma_m = G_{\text{sat}} G_{\text{ut}} L_{\text{t}} \lambda^2/(4\pi D_m)^2$, where G_{sat} is the satellite antenna elements gains, G_{ut} is the user antenna elements gains, and the distance between the mth legitimate user and satellite are represented as D_m [50]. The other detail of simulation setup parameters for LEO satellite systems are listed in Table III .

First of all, we investigate the convergence of the iterative algorithms. The curves in Fig. 3 show the WCSSR of each iterative step under different parameter settings, respectively. The results indicate that the objectives output by Algorithm 1 and Algorithm 2 increase monotonically and converge within a short number of iterations. Therefore, the algorithms we proposed exhibit fast convergence performance.

The WCSSR against the number of satellite antennas under different system settings is described in Figs. 4 and 5. It can be found that using more antennas can significantly improve the WCSSR, especially when the number of antennas is small, the growth trend is more obvious. Therefore, we

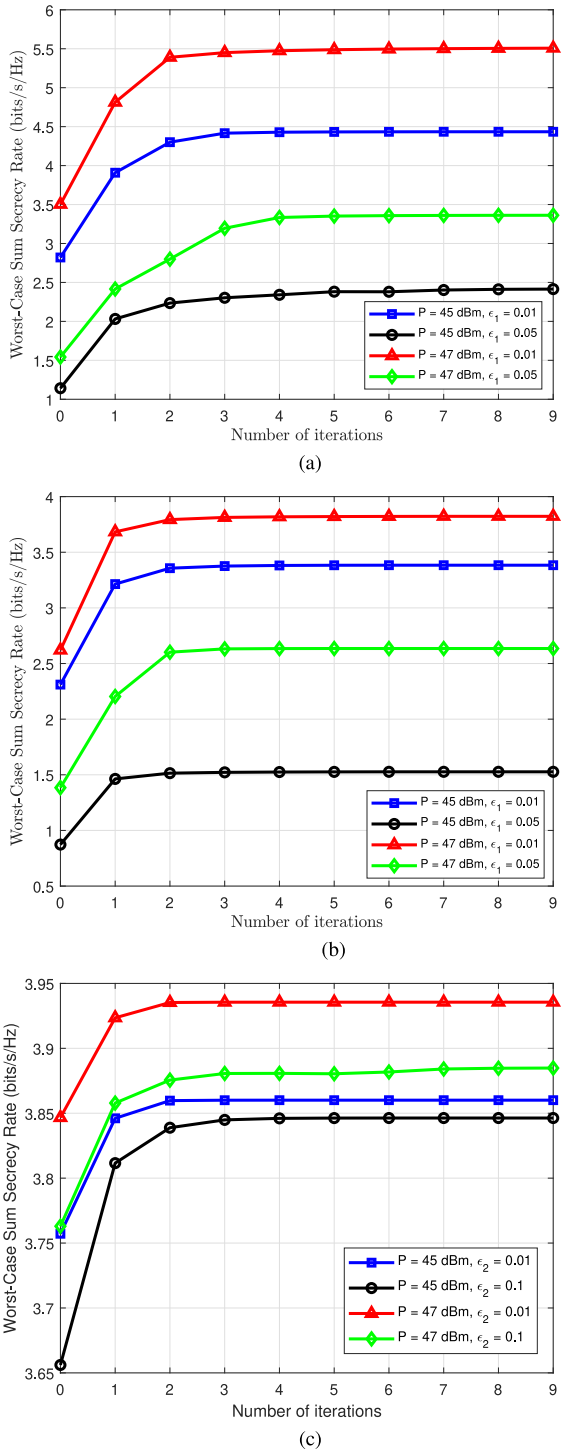


Fig. 3. Convergence behavior of the proposed approaches versus the number of iterations for different system setup parameters. (a) Algorithm 1 with $\rho = 15^\circ$. (b) Algorithm 1 with $\rho = 40^\circ$. (c) Algorithm 2.

can increase the number of satellite antennas to improve the security performance of the satellite systems. In addition, we can observe that different epsilon values will affect the growth trend of the WCSSR. When the epsilon value is larger, the growth trend of the WCSSR is slower, which means the system performance gain is smaller. Therefore, we can achieve a nearly optimal WCSSR by controlling

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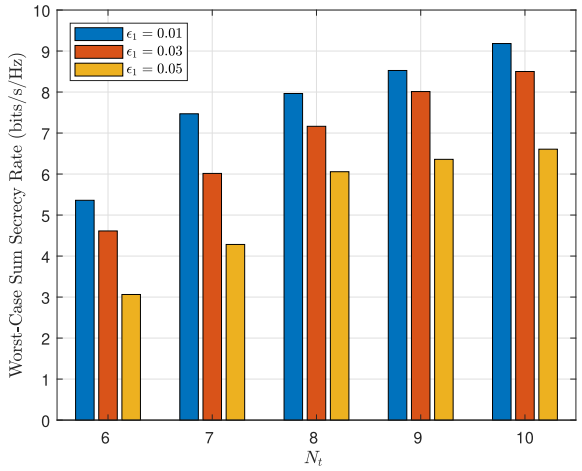


Fig. 4. WCSSR of Algorithm 1 versus the number of beams N_{t} with $P = 50$ dBm, and $\sigma = 15^{\circ}$.

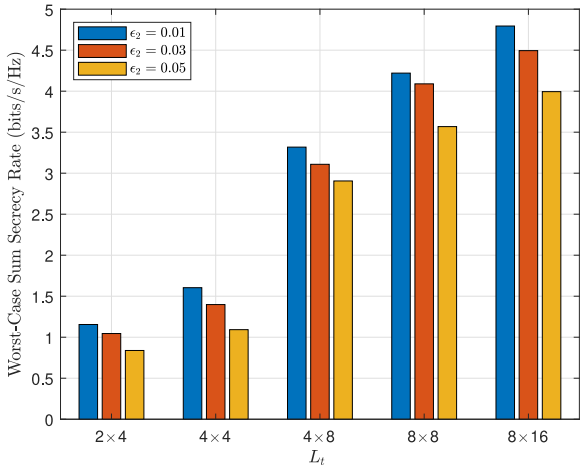


Fig. 5. WCSSR of Algorithm 2 versus the number of antennas L_{t} with $P = 50$ dBm, $[\varphi_L/\theta_L, \varphi_U/\theta_U] = [-5^{\circ}, 5^{\circ}]$.

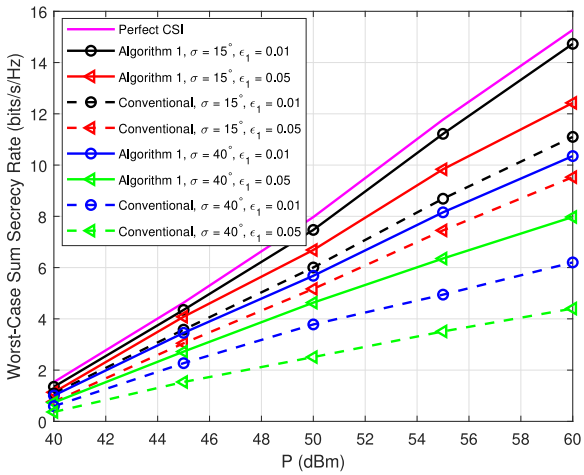


Fig. 6. Comparison between the proposed Algorithm 1 and conventional approach in terms of the worst-case secrecy rate.

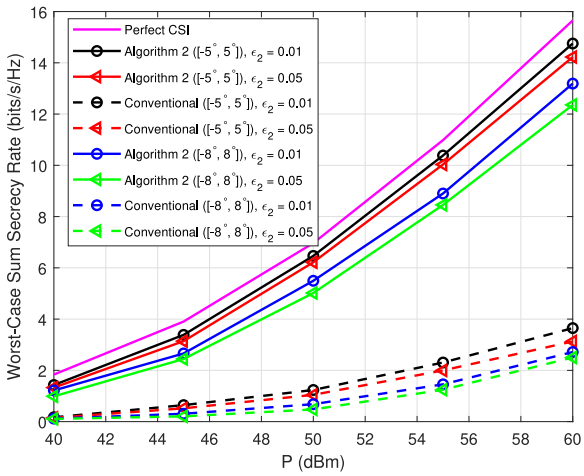


Fig. 7. Comparison between the proposed Algorithm 2 and conventional approach in terms of the worst-case secrecy sum rate.

the number of antennas and epsilon values to obtain better security and robustness of the GEO and LEO systems.

Fig. 6 depicts the comparison of the system security performance between our proposed secure transmission Algorithm 1 and the conventional approach. The conventional approach is to directly adopt the outdated CSI, which means that the estimated channel is used as the actual channel without considering the channel uncertainty of the user and the eavesdropper [13]. As observed, the proposed robust Algorithm 1 can obtain relatively higher WCSSR than the traditional approach. Moreover, the numerical results illustrate that the system security performance gains are significantly improved and the robustness is superior when the phase error variances are decreased. It also notes that with the increase of the transmission power, the corresponding WCSSR also increases. The Algorithm 1 has better security performance when the power is larger. In addition, the security performance of the GEO satellite is also related to the value of ϵ_1. The larger value of the ϵ_1, which means the greater the uncertainty of the eavesdropper CSI, the worse WCSSR would be acquired. Hence, in order to achieve satisfactory security performance in practical applications, the channel phase error σ and eavesdropper position error ϵ_1 should be strictly controlled under a tight region.

Similarly, the security performance curves of Algorithm 2 for different angular deviation uncertainty, eavesdropper position uncertainty, and power constraints are illustrated in Fig. 7. As excepted, when the power constraints increase and the eavesdropper channel uncertainly bound decreases, the WCSSR becomes larger, which means the system security performance is better. Therefore, we need to control the transmit power to a certain extent to obtain a higher system security performance gain. In addition, the Algorithm 2 is compared with the conventional approach in Fig. 7. Assuming that the angle deviation exhibits the uniform distribution in $[-5^{\circ}, 5^{\circ}]$ and $[-8^{\circ}, 8^{\circ}]$, respectively. We can observe that when the power constraint is small, the WCSSR of the two schemes is close, and the security performance of the robust scheme is more stable than the traditional one when the power constraint increases, which proves the robustness and effectiveness of our proposed secure transmission scheme.

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In addition, the simulation results also indicate that the WCSSR decreases as the uniform distribution range increases, that is, the greater the angular deviation uncertainty of the user, the smaller the user rate that can be obtained.

V. CONCLUSION

In this article, robust secure transmission schemes in GEO and LEO satellite scenarios have been proposed to improve the PLS of satellite communications. By considering the channel uncertainty of legitimate users and eavesdroppers, we maximized the WCSSR of GEO and LEO satellite systems, which guaranteed the constraint of transmit power for legitimate users. For the nonconvexity and intractability of the resulting optimization problems, an effective algorithm framework was proposed to tackle the problems. The SDR, CCP, Taylor expansion, and S-Procedure were employed to convert the original problem into the convex optimization subproblems that can be addressed iteratively. Finally, simulation results demonstrated the the effectiveness and superiority of our proposed secure transmission algorithms and the performance gain of the system can be significantly improved when CSI uncertainty are strictly controlled under a tight region.

APPENDIX

A. Proof of Proposition 1

We introduce the auxiliary variables $\{\beta_{3,k}\}_{k=1}^{K}$ and $\{\beta_{4,k}\}_{k=1}^{K}$ into constraint (24c), which after some mathematical computation, can be conservatively approximated as

$$\beta_{2,k} + \log_2 \beta_{3,k} - \log_2 \beta_{4,k} \geq 0 \tag{67a}$$

$$\log_2 \left(\mathrm{Tr}\left(\sum_{\ell \neq k} \mathbf{h}_e^H \mathbf{W}_\ell \mathbf{h}_e \right) + \sigma_g^2 \right) \geq \log_2 \beta_{3,k} \tag{67b}$$

$$\log_2 \left(\sum_{k=1}^{K} \mathbf{h}_e^H \mathbf{W}_k \mathbf{h}_e + \sigma_g^2 \right) \leq \log_2 \beta_{4,k}. \tag{67c}$$

For the constraint (66a), we utilize the first-order Taylor series expansion to transformed it into

$$\beta_{2,k} + \log_2 \beta_{3,k} - \log_2 \beta_{4,k}^{(\lambda)} - \frac{\beta_{4,k} - \beta_{4,k}^{(\lambda)}}{\beta_{4,k}^{(\lambda)} \ln 2} \geq 0. \tag{68}$$

Then, by considering the channel uncertainties in (11), we rewrite constraint (67b) as

$$\begin{cases} \forall \Delta\mathbf{h}_e : \Delta\mathbf{h}_e^H \Delta\mathbf{h}_e - \epsilon_1^2 \leq 0 \\ -\Delta\mathbf{h}_e^H \mathbf{F}_1 \Delta\mathbf{h}_e - 2\mathrm{Re}\left\{ \hat{\mathbf{h}}_e^H \mathbf{F}_1 \Delta\mathbf{h}_e \right\} - \vartheta_1 \leq 0 \end{cases} \tag{69}$$

where $\vartheta_1 = \hat{\mathbf{h}}_e^H \mathbf{F}_1 \hat{\mathbf{h}}_e - \beta_{3,k} + \sigma_g^2$. Applying the S-Procedure in (69), we can transform it into the form of (35b). To deal with the constraint (67c), by employing the similar approaches, we can acquire

$$\begin{cases} \forall \Delta\mathbf{h}_e : \Delta\mathbf{h}_e^H \Delta\mathbf{h}_e - \epsilon_1^2 \leq 0 \\ \Delta\mathbf{h}_e^H \mathbf{F}_2 \Delta\mathbf{h}_e + 2\mathrm{Re}\left\{ \hat{\mathbf{h}}_e^H \mathbf{F}_2 \Delta\mathbf{h}_e \right\} + \vartheta_2 \leq 0 \end{cases} \tag{70}$$

where $\vartheta_2 = \hat{\mathbf{h}}_e^H \mathbf{F}_2 \hat{\mathbf{h}}_e - \beta_{4,k} + \sigma_g^2$. Then, according to the S-Procedure, (70) can be converted as the constraint (35c). This concludes the proof.

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REFERENCES

[1] J. Ye, S. Dang, B. Shihada, and M.-S. Alouini, "Space-air-ground integrated networks: Outage performance analysis," *IEEE Trans. Wireless Commun.*, vol. 19, no. 12, pp. 7897–7912, Dec. 2020.

[2] L. You et al., "Beam squint-aware integrated sensing and communications for hybrid massive MIMO LEO satellite systems," *IEEE J. Sel. Areas Commun.*, to be published, doi: 10.1109/JSAC.2022.3196114.

[3] F. Alagoz and G. Gur, "Energy efficiency and satellite networking: A holistic overview," *Proc. IEEE*, vol. 99, no. 11, pp. 1954–1979, Nov. 2011.

[4] M. Á. Vázquez et al., "Precoding in multibeam satellite communications: Present and future challenges," *IEEE Wireless Commun.*, vol. 23, no. 6, pp. 88–95, Dec. 2016.

[5] X. Qiang et al., "Hybrid A/D precoding for downlink massive MIMO in LEO satellite communications," in *Proc. IEEE Int. Conf. Commun.*, Montreal, QC, Canada, Jul. 2021, pp. 1–6.

[6] L. You et al., "Hybrid analog/digital precoding for downlink massive MIMO LEO satellite communications," *IEEE Trans. Wireless Commun.*, vol. 21, no. 8, pp. 5962–5976, Aug. 2022.

[7] A. D. Wyner, "The wire-tap channel," *Bell System Tech. J.*, vol. 54, no. 8, pp. 1355–1387, Oct. 1975.

[8] S. Spinsante, F. Chiaraluce, and E. Gambi, "Evaluation of AES-based authentication and encryption schemes for telecommand and telemetry in satellite applications," in *Proc. SpaceOps Conf.*, 2006, pp. 1–15.

[9] A. Mukherjee, S. A. A. Fakoorian, J. Huang, and A. L. Swindlehurst, "Principles of physical layer security in multiuser wireless networks: A survey," *IEEE Commun. Surv. Tuts.*, vol. 16, no. 3, pp. 1550–1573, Sep. 2014.

[10] D. Kapetanovic, G. Zheng, and F. Rusek, "Physical layer security for massive MIMO: An overview on passive eavesdropping and active attacks," *IEEE Commun. Mag.*, vol. 53, no. 6, pp. 21–27, Jun. 2015.

[11] A. Gharanjik, M. R. B. Shankar, P.-D. Arapoglou, M. Bengtsson, and B. Ottersten, "Precoding design and user selection for multibeam satellite channels," in *Proc. IEEE 21st Int. Workshop Signal Process. Adv. Wirel. Commun.*, Stockholm, Sweden, 2015, pp. 420–424.

[12] K. Cumanan, Z. Ding, B. Sharif, G. Y. Tian, and K. K. Leung, "Secrecy rate optimizations for a MIMO secrecy channel with a multiple-antenna eavesdropper," *IEEE Trans. Veh. Technol.*, vol. 63, no. 4, pp. 1678–1690, May 2013.

[13] A. Gharanjik, M. R. B. Shankar, P.-D. Arapoglou, M. Bengtsson, and B. Ottersten, "Robust precoding design for multibeam downlink satellite channel with phase uncertainty," in *Proc. IEEE Int. Conf. Acoust., Speech, Signal Process.*, Brisbane, Queensland, Australia, 2015, pp. 3083–3087.

[14] X. Zhang, J. Wang, C. Jiang, C. Yan, Y. Ren, and L. Hanzo, "Robust beamforming for multibeam satellite communication in the face of phase perturbations," *IEEE Trans. Veh. Technol.*, vol. 68, no. 3, pp. 3043–3047, Mar. 2019.

[15] L. You, A. Liu, W. Wang, and X. Gao, "Outage constrained robust multigroup multicast beamforming for multi-beam satellite communication systems," *IEEE Wireless Commun. Lett.*, vol. 8, no. 2, pp. 352–355, Apr. 2019.

[16] W. Wang, Y. Tong, L. Li, A.-A. Lu, L. You, and X. Gao, "Near optimal timing and frequency offset estimation for 5G integrated LEO satellite communication system," *IEEE Access*, vol. 7, pp. 113298–113310, Aug. 2019.

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[17] Z. Lin, M. Lin, J. Ouyang, W.-P. Zhu, A. D. Panagopoulos, and M.-S. Alouini, "Robust secure beamforming for multibeam satellite communication systems," *IEEE Trans. Veh. Technol.*, vol. 68, no. 6, pp. 6202–6206, Jun. 2019.
[18] L. You, J. Wang, W. Wang, and X. Gao, "Secure multicast transmission for massive MIMO with statistical channel state information," *IEEE Signal Process. Lett.*, vol. 26, no. 6, pp. 803–807, Jun. 2019.
[19] J. Wang, J. Lee, F. Wang, and T. Q. S. Quek, "Jamming-aided secure communication in massive MIMO rician channels," *IEEE Trans. Wireless Commun.*, vol. 14, no. 12, pp. 6854–6868, Dec. 2015.
[20] G. Zheng, P.-D. Arapoglou, and B. Ottersten, "Physical layer security in multibeam satellite systems," *IEEE Trans. Wireless Commun.*, vol. 11, no. 2, pp. 852–863, Feb. 2012.
[21] J. Wang and D. P. Palomar, "Worst-case robust MIMO transmission with imperfect channel knowledge," *IEEE Trans. Signal Process.*, vol. 57, no. 8, pp. 3086–3100, Aug. 2009.
[22] W. Wang, X. Chen, L. You, X. Yi, and X. Gao, "Artificial noise assisted secure massive MIMO transmission exploiting statistical CSI," *IEEE Commun. Lett.*, vol. 23, no. 12, pp. 2386–2389, Dec. 2019.
[23] J. Huang and A. L. Swindlehurst, "Robust secure transmission in MISO channels based on worst-case optimization," *IEEE Trans. Signal Process.*, vol. 60, no. 4, pp. 1696–1707, Apr. 2011.
[24] W. Lu, K. An, and T. Liang, "Robust beamforming design for sum secrecy rate maximization in multibeam satellite systems," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 55, no. 3, pp. 1568–1572, Jun. 2019.
[25] X. Wang, K. Wang, and X.-D. Zhang, "Secure relay beamforming with imperfect channel side information," *IEEE Trans. Veh. Technol.*, vol. 62, no. 5, pp. 2140–2155, Jun. 2013.
[26] G. Zheng, S. Chatzinotas, and B. Ottersten, "Generic optimization of linear precoding in multibeam satellite systems," *IEEE Trans. Wireless Commun.*, vol. 11, no. 6, pp. 2308–2320, Jun. 2012.
[27] C. Caini, G. E. Corazza, G. Falciasecca, M. Ruggieri, and F. Vatalaro, "A spectrum- and power-efficient EHF mobile satellite system to be integrated with terrestrial cellular systems," *IEEE J. Sel. Areas Commun.*, vol. 10, no. 8, pp. 1315–1325, Oct. 1992.
[28] M. Díaz, N. Courville, C. Mosquera, G. Liva, and G. Corazza, "Non-linear interference mitigation for broadband multimedia satellite systems," in *Proc. Int. Workshop Satell. Space Commun.*, 2007, pp. 61–65.
[29] W. Wang et al., "Resource efficiency optimization for robust beamforming in multi-beam satellite communications," *IEEE Trans. Veh. Technol.*, vol. 70, no. 7, pp. 6958–6968, Jul. 2021.
[30] V. Joroughi, M. Á. Vázquez, and A. I. Pérez-Neira, "Generalized multicast multibeam precoding for satellite communications," *IEEE Trans. Wireless Commun.*, vol. 16, no. 2, pp. 952–966, Feb. 2017.
[31] T. Maseng and P. Bakken, "A stochastic dynamic model of rain attenuation," *IEEE Trans. Wireless Commun.*, vol. 29, no. 5, pp. 660–669, May 1981.
[32] G. Taricco, "Linear precoding methods for multi-beam broadband satellite systems," in *Proc. 20th Eur. Wireless Conf.*, Barcelona, Spain, May 2014, pp. 1–6.
[33] Y. Yan, K. An, B. Zhang, W.-P. Zhu, G. Ding, and D. Guo, "Outage-constrained robust multigroup multicast beamforming for satellite-based Internet of Things coexisting with terrestrial networks," *IEEE Internet Things J.*, vol. 8, no. 10, pp. 8159–8172, May 2021.
[34] W. Wang, A. Liu, Q. Zhang, L. You, X. Gao, and G. Zheng, "Robust multigroup multicast transmission for frame-based multi-beam satellite systems," *IEEE Access*, vol. 6, pp. 46074–46083, Aug. 2018.
[35] M. Shao and W.-K. Ma, "A simple way to approximate average robust multiuser MISO transmit optimization under covariance-based CSIT," in *Proc. IEEE Int. Conf. Acoust., Speech, Signal Process.*, New Orleans, LA, USA, Mar. 2017, pp. 3504–3508.
[36] Q. Zhang, S. Jin, K.-K. Wong, H. Zhu, and M. Matthaiou, "Power scaling of uplink massive MIMO systems with arbitrary-rank channel means," *IEEE J. Sel. Topics Signal Process.*, vol. 8, no. 5, pp. 966–981, Oct. 2014.
[37] L. Gao, J. Ma, L. You, C. Pan, W. Wang, and X. Gao, "Robust energy-efficient multigroup multicast beamforming for multi-beam satellite communications," in *Proc. IEEE Int. Conf. Commun.*, Dublin, Ireland, Jul. 2020, pp. 1–6.
[38] T. Lipp and S. Boyd, "Variations and extension of the convex–concave procedure," *J. Optim. Eng.*, vol. 17, no. 2, pp. 263–287, Jun. 2016.
[39] A. Zappone, P.-H. Lin, and E. Jorswieck, "Energy efficiency of confidential multi-antenna systems with artificial noise and statistical CSI," *IEEE J. Sel. Topics Signal Process.*, vol. 10, no. 8, pp. 1462–1477, Dec. 2016.
[40] C. Sun, X. Gao, and Z. Ding, "BDMA in multicell massive MIMO communications: Power allocation algorithms," *IEEE Trans. Signal Process.*, vol. 65, no. 11, pp. 2962–2974, Jun. 2017.
[41] W. Wu, X. Gao, Y. Wu, and C. Xiao, "Beam domain secure transmission for massive MIMO communications," *IEEE Trans. Veh. Technol.*, vol. 67, no. 8, pp. 7113–7127, Aug. 2018.
[42] C. Sun, X. Gao, J. Wang, Z. Ding, and X.-G. Xia, "Beam domain massive MIMO for optical wireless communications with transmit lens," *IEEE Trans. Commun.*, vol. 67, no. 3, pp. 2188–2202, Mar. 2019.
[43] I. Pólik and T. Terlaky, "Interior point methods for nonlinear optimization," in *Nonlinear Optimization*. Berlin, Germany: Springer, 2010, pp. 215–276.
[44] S. Wang and B. Wang, "Robust secure transmit design in MIMO channels with simultaneous wireless information and power transfer," *IEEE Signal Process. Lett.*, vol. 22, no. 11, pp. 2147–2151, Aug. 2015.
[45] A. Adhikary, J. Nam, J.-Y. Ahn, and G. Caire, "Joint spatial division and multiplexing-the large-scale array regime," *IEEE Trans. Inf. Theory*, vol. 59, no. 10, pp. 6441–6463, Oct. 2013.
[46] L. You, K.-X. Li, J. Wang, X. Gao, X.-G. Xia, and B. Ottersten, "Massive MIMO transmission for LEO satellite communications," *IEEE J. Sel. Areas Commun.*, vol. 38, no. 8, pp. 1851–1865, Aug. 2020.
[47] R. Shafin, L. Liu, Y. Li, A. Wang, and J. Zhang, "Angle and delay estimation for 3-D massive MIMO/FD-MIMO systems based on parametric channel modeling," *IEEE Trans. Wireless Commun.*, vol. 16, no. 8, pp. 5370–5383, Aug. 2017.
[48] A. Adnane, Z. A. Foitih, M. A. S. Mohammed, and A. Bellar, "Real-time sensor fault detection and isolation for LEO satellite attitude estimation through magnetometer data," *Adv. Space Res.*, vol. 61, no. 4, pp. 1143–1157, Dec. 2018.
[49] D. Cilden-Guler, M. Raitoharju, R. Piche, and C. Hajiyev, "Nanosatellite attitude estimation using Kalman-type filters with non-Gaussian noise," *Aerosp. Sci. Technol.*, vol. 92, pp. 66–76, Sep. 2019.
[50] K.-X. Li et al., "Downlink transmit design for massive MIMO LEO satellite communications," *IEEE Trans. Commun.*, vol. 70, no. 2, pp. 1014–1028, Feb. 2022.
[51] J. Zhu, R. Schober, and V. K. Bhargava, "Linear precoding of data and artificial noise in secure massive MIMO systems," *IEEE Trans. Wireless Commun.*, vol. 15, no. 3, pp. 2245–2261, Mar. 2016.



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